UK RAILWAY TICKET SALES ANALYSIS

PROJECT DOCUMENTATION

Project Planning | Literature Review | Requirements Gathering

TEAM 4

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Contents

1. Executive Summary
2. Project Proposal
3. Project Plan and Timeline
4. Task Assignment and Team Roles
5. Risk Assessment and Mitigation
6. Key Performance Indicators
7. Literature Review
8. Feedback, Evaluation, Improvements, and Grading
9. Requirements Gathering
10. Stakeholder Analysis
11. User Stories and Use Cases
12. Functional Requirements
13. Non-functional Requirements
14. Scope, Assumptions, and Limitations
15. Conclusion

1. Executive Summary

This project examines UK railway ticket sales and journey performance using transaction-level data. Each record is expected to connect ticket information—such as price, ticket type, class, railcard and purchase channel—with journey outcomes, including whether a service was on time, delayed or cancelled. Where available, the data also records delay reasons and whether a refund was requested.

The purpose is to turn raw records into reliable, understandable evidence that can support operational, pricing and customer-service decisions. The work will begin with data inspection and cleaning, continue through exploratory analysis and carefully scoped predictive modelling, and finish with a dashboard and written report. The dashboard is intended to let stakeholders explore performance by route, station, date, ticket category and purchase characteristics.

The project is organised into three connected documentation areas: project planning and management, a literature review, and requirements gathering. Planning defines the work, responsibilities, schedule and measures of success. The literature review places the project in the context of existing work on railway delays, compensation and ticket pricing. Requirements gathering defines the people who may use the results, the questions they need answered, and the expected functions and quality of the analytical workflow.

A central principle is that conclusions must match what the dataset can actually show. Refund requests, for example, record passenger behaviour and should not be treated as a complete count of passengers eligible for compensation. Similarly, findings from this dataset should not automatically be generalised to the entire UK railway network.

Project at a glance

Item	Description
Project title	UK Railway Ticket Sales and Journey Performance Analysis
Project type	Data analytics, business intelligence and exploratory predictive modelling
Primary tool	Power BI for data transformation, analysis and dashboard presentation
Supporting methods	Data quality checks, descriptive statistics, visual analysis and model evaluation
Main outputs	Cleaned dataset, data-quality log, analytical findings, dashboard, report and presentation
Primary users	Rail operations managers, pricing analysts, customer-relations teams, project team and lecturer

2. Project Proposal

2.1 Background and rationale

Railway performance is shaped by several connected factors. Passengers choose tickets with different prices, classes and conditions, while operators need to understand service reliability and the reasons behind disruption. When a journey is delayed or cancelled, a passenger may or may not request a refund. Looking at these fields together makes it possible to explore questions that would be difficult to answer from a single summary of train punctuality or ticket sales.

The project uses the available ticket and journey records as its evidence base. Rather than assuming that every field is complete or error-free, the team will inspect the data structure, confirm the meaning of each field, identify missing or inconsistent values, and document any decisions made during preparation.

2.2 Aim

The overall aim is to produce a trustworthy and accessible analysis of UK railway ticket sales and journey performance, identifying useful patterns in delays, pricing and refunds and presenting the findings in a clear dashboard and report.

2.3 Objectives

- Inspect the supplied dataset and confirm that expected fields, formats and values are present.
- Clean date and time fields, identify duplicates, assess missing values and flag records that may be logically inconsistent.
- Create useful analytical fields, such as journey delay minutes, booking lead time, route, day, hour and peak/off-peak indicator, where the source data supports them.
- Describe punctuality, delays and cancellations by route, station, time period and relevant ticket attributes.
- Compare ticket prices and sales patterns by ticket type, class, railcard, purchase channel and booking lead time.
- Analyse refund-request patterns among delayed or cancelled journeys without treating a request as proof of eligibility.
- Explore a baseline and at least one predictive model for delayed versus on-time journeys, using only information available before travel.
- Evaluate models using suitable test-set metrics, including precision, recall and F1-score where applicable.
- Develop a clear dashboard with useful KPIs, filters and labelled visualisations.
- Prepare a final report and presentation that explain the method, findings, limitations and recommendations.

2.4 Project scope

The project covers the data preparation, descriptive analysis, dashboard development and a limited predictive exercise described in the requirements. It is an analytical workflow rather than a live railway control system. The work will not directly control train services, issue refunds, change ticket prices or provide real-time alerts unless these capabilities are separately specified and implemented.

Included

- Data loading, validation, cleaning and documentation.
- Exploratory analysis of journey status, delays, delay reasons, pricing and refund requests.
- Comparisons across routes, stations, dates, ticket types, classes, railcards and purchase channels where fields are available.

- A baseline and at least one model to assess delay risk, subject to data suitability.
- Power BI dashboard design, interactive filters and reporting.
- Final documentation, recommendations and presentation.

Outside the current scope

- Live integration with railway operational systems or live train feeds.
- Automatic refund processing or passenger-level decisions.
- Claims that the sample represents every UK railway journey.
- Use of post-journey information to predict a delay before the journey.
- External weather or train-load data unless the team obtains and documents suitable sources.

2.5 Expected value

The project is expected to provide a structured view of where disruption is concentrated, how ticket prices vary across available categories, and how refund requests relate to journey outcomes. It can also demonstrate whether a simple delay-risk model adds useful information beyond a baseline. These findings may help stakeholders decide where further investigation is warranted; they should be treated as evidence for discussion, not as proof of causation.

3. Project Plan and Timeline

The schedule below is a proposed four-week plan. The team should adjust dates to match the course calendar, dataset availability and lecturer feedback. Each phase ends with a tangible deliverable so that progress can be checked before the next phase begins.

Week / phase	Main activities	Milestone	Deliverables
Week 1 — Data preparation	Inspect fields; profile data quality; clean dates, times, labels and duplicates; document missing and invalid records; create supported derived fields.	Dataset is ready for analysis and major data-quality decisions are documented.	Cleaned dataset; preprocessing notes; data-quality log.
Week 2 — Analysis and modelling	Define stakeholder questions; calculate descriptive measures; analyse delays, pricing and refunds; establish a baseline; test a suitable model if data supports it.	Analysis questions and initial findings are agreed by the team.	Analysis question list; summary tables/charts; model evaluation notes.
Week 3 — Dashboard	Design pages and KPIs; create measures; build visuals and filters; validate calculations; collect team feedback and revise layout.	Dashboard review completed and key numbers checked against analysis outputs.	Power BI dashboard; KPI definitions; validation checklist.
Week 4 — Finalisation	Write the report; document methods and limitations; prepare presentation; rehearse delivery; apply lecturer feedback where time permits.	Final submission package is complete.	Final report; presentation; project files and supporting documentation.

3.1 Gantt-style schedule

Activity	W1	W2	W3	W4
Project scope and question definition	■	■		
Data profiling and cleaning	■			
Data-quality documentation	■	■		
Exploratory analysis		■		
Predictive modelling and evaluation		■	■	
Dashboard design and development			■	
Dashboard validation and revision			■	■
Report writing			■	■
Presentation preparation and rehearsal				■
Final review and submission				■

Legend: ■ = planned work period. Some activities overlap deliberately; for example, report notes should be maintained throughout the project rather than written entirely at the end.

3.2 Milestones and acceptance checks

Milestone	Acceptance check
M1 — Data ready	Expected fields checked; data types corrected; key cleaning decisions and unresolved issues recorded.
M2 — Analysis agreed	Questions are answerable from available fields and definitions for key measures are documented.
M3 — Dashboard reviewed	Visuals reconcile with summary outputs; filters work; titles and units are clear.
M4 — Final package ready	Report, presentation, cleaned data and project files are checked and submitted.

3.3 Resource allocation

The core resources are the supplied dataset, Power BI Desktop, a shared project folder, a versioned copy of working files where practical, and the time contributed by each team member. A spreadsheet or document can be used to track decisions and issues. If modelling is completed outside Power BI, the team should record the tool, version, data split, random seed and evaluation metrics so that the work can be reproduced.

4. Task Assignment and Team Roles

Names are intentionally left blank so the team can assign responsibilities. Roles can be combined for a small team, but every major deliverable should have one clearly accountable owner. All members should review the final results, and the person responsible for a task should not be the only person checking its quality.

Role / workstream	Main responsibilities	Assigned member
Project coordinator	Maintain the plan, meeting notes, milestone tracker and communication with the lecturer; follow up on blockers; coordinate final submission.	Ahmed Adel Mesbah Ahmed
Data preparation lead	Inspect fields; perform cleaning and transformations; document missing values, duplicates, anomalies and derived fields; maintain the clean-data version.	Khaled Abdelhamid Mahmoud Rizk
Analysis lead	Develop analysis questions; define KPIs; calculate descriptive measures; analyse delays, pricing, refunds and route/time patterns.	Ahmed Mohamed Elsayed Tolba
Modelling lead	Define the prediction target; prevent data leakage; build a baseline and candidate model; evaluate on held-out data; document limitations.	Yassa Ezzat Magdy Abdo
Power BI / dashboard lead	Design the report pages; build measures and visualisations; add filters; maintain consistent layout and readable labels.	Hussein Mohamed Hussein Ibrahim
Documentation and presentation lead	Combine the report sections; maintain references; prepare slides; ensure findings and recommendations match the evidence.	Ahmed Mohamed Elsayed Tolba

5. Risk Assessment and Mitigation Plan

Risk management is intended to identify issues early enough that the team can respond without compromising the quality of the analysis. Ratings below are initial estimates; the team should update them when it has inspected the actual dataset and confirmed the course deadlines.

Risk	Likelihood	Impact	Mitigation / response
Missing, duplicated or inconsistent records	Medium	High	Profile the dataset at the start; preserve raw data; log changes; quantify exclusions; validate important transformations.
Incorrect date/time handling, especially journeys crossing midnight	Medium	High	Test known examples; check arrival/departure logic; document assumptions and manually verify a sample.
Required fields are absent or have different meanings	Medium	High	Map source fields to requirements; confirm definitions with lecturer; revise questions rather than invent unsupported fields.
Delay model leaks information unavailable before travel	Medium	High	Exclude actual arrival, delay reason, final journey status and refund request from pre-journey predictors where they reveal the outcome.
Class imbalance makes accuracy misleading	Medium	Medium	Compare with a baseline and report precision, recall and F1-score; describe the minority class clearly.
Forecast or model performs poorly on unseen data	Medium	Medium	Use a held-out test set, compare with a simple baseline, report limitations and avoid overstating predictive value.
Dashboard contains inconsistent measures	Medium	High	Maintain a KPI dictionary; reconcile totals to the cleaned data; perform peer review and test filters.
Schedule slippage or uneven workload	Medium	Medium	Use weekly milestones, assign one owner per task, track progress and reduce optional scope before sacrificing core validation.
File loss or conflicting versions	Low/Medium	Medium	Keep backups, use clear versioned filenames and agree on a shared location and file ownership.
Privacy or inappropriate interpretation of group results	Low/Medium	High	Share aggregate results only, avoid identifying individuals and describe railcard-group findings cautiously.
Presentation or report is incomplete	Medium	Medium	Reserve time in Week 4; use a submission checklist; rehearse the presentation and check all references.

6. Key Performance Indicators (KPIs)

Project KPIs measure whether the work is complete, reliable and useful. They are different from railway business measures such as on-time rate or refund-request rate. The targets below are proposed project-quality targets, not claims about actual railway performance. The team should adjust them if the course brief specifies different thresholds.

KPI	Definition / target	How it will be checked
Data completeness review	All expected fields reviewed; missing values quantified and explained.	Data-quality summary and field-level checks.
Duplicate handling	Duplicate Transaction IDs identified and their treatment documented.	Duplicate report before and after cleaning.
Data validation	At least 30 records spot-checked against the raw data with no unexplained transformation errors.	Recorded sample checks and correction log.
Reproducibility	Another team member can repeat the documented workflow and obtain the same key totals.	Peer rerun and reconciliation.
Dashboard responsiveness	Filters refresh in under 3 seconds on the available test machine, where feasible.	Timed filter tests.
Pipeline execution	Full workflow completes in under 5 minutes on a normal laptop, where feasible.	Timed run and note of hardware/software.
Visual clarity	Every chart has a descriptive title, units where needed and a clear interpretation.	Dashboard review checklist.
Model evaluation	Model is evaluated on unseen test data and compared with a simple baseline.	Recorded split, metrics and comparison.
Leakage control	No predictor contains information that would only be known after the journey when predicting before travel.	Feature review and modelling notes.
Deliverable completion	All agreed deliverables are submitted by the course deadline.	Milestone tracker and final checklist.
Stakeholder usability	A reviewer can locate key KPIs and use dashboard filters without step-by-step help.	Short peer walkthrough and feedback.

6.1 Railway performance measures to consider

Measure	Suggested definition	Interpretation caution
On-time rate	On-time journeys divided by journeys with a usable outcome status.	Confirm the source definition of on time and the denominator.
Delay rate	Delayed journeys divided by journeys with a usable outcome status.	Keep cancelled journeys separate unless the chosen definition explicitly combines them.
Cancellation rate	Cancelled journeys divided by journeys with a usable outcome status.	State how missing or invalid status records are treated.
Average delay minutes	Mean delay duration among delayed journeys with valid delay values.	Do not silently treat missing delay duration as zero.
Refund-request rate	Journeys with a refund request divided by the stated eligible analysis group.	A request is not proof of eligibility; the dataset may not record eligibility.
Average ticket price	Mean ticket price for the selected population and filters.	Check currency, outliers, refunds and whether records represent tickets or transactions.
Booking lead time	Journey date/time minus purchase date/time, under the documented convention.	Handle date/time issues and negative lead times explicitly.

7. Literature Review

7.1 Purpose and focus

This review considers work relevant to the project's UK rail ticket and journey dataset. The supplied literature review focuses on four areas: delay analysis and prediction, delay compensation and refunds, ticket pricing, and analytical methods. The review is a concise overview rather than a full systematic review, and some referenced papers were available only through their abstracts. Industry claims are not independently verified in the supplied review.

7.2 Delay analysis and prediction

The supplied review describes several studies that use machine learning to predict railway delays. Taleongpong et al. (2022) examine reactionary delays, dwell time and travel time on the British network. A University of East Anglia study combines multiple algorithms and tests them on a second dataset from another operator. Wang, Guo and Zhang (2024) apply a deep-learning approach to two UK railway lines. A study by Larson (2020) is noted as a useful caution: predicting the size of a delay does not necessarily explain where or why it happened. The review also mentions an industry tool used by LNER to warn station staff about likely delays.

For this project, these studies support careful exploration of delay patterns and a modest predictive exercise. A model should be compared with a baseline and tested on data it has not seen. The analysis should distinguish between explaining historical delay reasons and predicting delay risk before a journey. These are different tasks and require different input fields.

7.3 Compensation and refund requests

The supplied review summarises Transport Focus (2016) and a Department for Transport survey (2020), both of which report that only a portion of eligible passengers claim compensation. It notes that the reported claim rate varies with the length of delay and that some passengers may not know they can claim. These findings matter because the dataset's Refund Request field records what passengers did, not whether they were entitled to compensation.

Accordingly, the project will analyse refund-request patterns among delayed or cancelled journeys and compare them across available fields. It will avoid treating the observed request rate as the true compensation eligibility rate. If eligibility is not present in the data, it cannot be inferred conclusively from the refund field alone.

7.4 Ticket pricing and purchase behaviour

The supplied review discusses Trainline's historical price-prediction tool and notes that this is an industry claim. It also cites Anciaes et al. (2019), whose work on fare complexity suggests that ticket type can influence rail demand. Since the dataset includes purchase date, journey date, ticket type and price, the project can examine price differences by ticket category and booking lead time. It can also compare sales across purchase channels and payment methods if these fields are populated and consistently defined.

7.5 Analytical methods and methodological cautions

The supplied review recommends CRISP-DM as a project process: understand the business problem, understand and prepare the data, model, evaluate and report. It also recommends exploratory data analysis before modelling. Random forests and gradient boosting, including XGBoost, are mentioned as potential methods for tabular data. The review warns that delayed journeys may be a minority class, so accuracy alone can give a misleading impression of model quality.

The project will use the methods that are appropriate for the available data, team skills and course requirements. It will report the target definition, input fields, data split and evaluation metrics.

Information only known after the journey—such as actual arrival time, the final reason for delay, refund request or final status—must not be used as an input to a model intended to predict delay before travel.

7.6 Research gaps and project response

Gap highlighted in supplied review	How this project responds
Some delay studies focus on train-level data rather than ticket and passenger-related attributes.	Compare outcomes across ticket type, class, railcard, purchase channel and route where available.
Prediction of delay size may not explain why a delay happened.	Rank recorded delay reasons and compare them across routes; describe these as observed associations.
Refund behaviour is often measured through surveys rather than transaction records.	Use transaction-level refund-request records to measure observed requests and compare them with journey outcomes.
Pricing research may not link ticket price and journey reliability.	Compare prices by ticket type and booking lead time and explore differences between on-time and delayed journeys.

7.7 Limitations of the literature

The supplied review is intentionally brief. Some claims are based on abstracts, survey summaries or industry publications, and not every source has the same level of independence. The project should therefore use the review to frame questions rather than treat every reported result as a guaranteed pattern in this dataset.

8. Feedback, Evaluation, Improvements, and Grading

8.1 Suggested improvements

The following improvements are consistent with the supplied literature review and requirements document. They are possible extensions rather than claims that the current scope already includes them.

- Add external data such as weather or train-load information if reliable, compatible data can be obtained and documented.
- Test models on a second time period or operator dataset to assess whether performance generalises.
- Improve dashboard filters so stakeholders can explore results by route, ticket type and date.
- Investigate differences between refund-request behaviour and journey outcomes while being explicit about the absence of eligibility data.
- Document the assumptions behind delay duration, missing values, date/time conversion and cancellation handling.
- Add model interpretation and error analysis so stakeholders understand where predictions work poorly as well as where they perform well.

9. Requirements Gathering

9.1 Project overview

The analytical workflow starts with ticket records and produces a cleaned dataset, analytical outputs, a dashboard and a report. Each record is described in the supplied requirements document as including purchase information, price, ticket type, class, railcard, route, journey outcome, delay reason where relevant, and refund-request information. The source document states that the expected dataset contains 18 fields; this must be checked against the actual file before work begins.

9.2 Requirement priorities

Requirements use three priority levels: Must, Should and Could. Must requirements define the core outcome expected for the project. Should requirements add useful capabilities when time and data allow. Could requirements are optional extensions and should not take priority over validation of the core results.

9.3 Data assumptions

- The full dataset is assumed to have the same structure as the sample; this must be verified when the full file is available.
- The meaning of each field should be confirmed from the dataset documentation or lecturer.
- A blank delay reason on an on-time journey may be expected rather than a data error.
- A record that appears illogical should be flagged and investigated before being removed.
- The dataset describes the supplied records and should not be treated as a representative sample of the entire UK network without further evidence.

9.4 Analytical questions

Area	Questions the project can investigate
Journey performance	What shares of journeys are on time, delayed or cancelled? How do these shares vary by route, station, date and time?
Delay reasons	Which recorded delay reasons are most frequent? Which are associated with the greatest recorded delay minutes?
Pricing	How does price vary by ticket type, class, railcard and route? Is price associated with booking lead time?

Purchases	How do purchase channel and payment method vary across available transaction records?
Refund requests	What proportion of delayed or cancelled journeys has a refund request recorded, and how does this vary by delay length and ticket attributes?
Prediction	Can a baseline and a suitable model identify journeys at higher risk of delay using information available before travel?
Dashboard use	Can a stakeholder filter results by date, route, ticket type, class, railcard, channel and status?

10. Stakeholder Analysis

Stakeholders are assessed by their likely influence on the project and their interest in the results. The ratings below are taken from the supplied requirements document and should be confirmed with the lecturer.

Stakeholder	Needs from the analysis	Influence	Interest
Train operator / operations managers	Where, when and why delays happen; worst-performing routes and stations.	High	High
Pricing and revenue analysts	Price differences by ticket type, class, railcard and booking time; purchase behaviour.	High	High
Customer relations / refund team	Share of delayed journeys with refund requests and patterns associated with requests.	Medium	High
Passengers (represented through the data)	Reliable services and fair treatment; understandable findings about performance.	Low	High
Regulators and passenger watchdogs	Accurate evidence about punctuality and compensation-related patterns.	High	Medium
Project team and lecturer	Clear scope, sound methods, reliable data and well-documented results.	High	High

10.1 Stakeholder engagement

- Consult the lecturer on scope, marking expectations and any ambiguous dataset definitions.
- Use operational and pricing questions to guide the selection of dashboard KPIs.
- Present refund-related findings in plain language and explain the limitations of the Refund Request field.
- Use a short peer review to check that non-technical users can interpret the dashboard without specialist support.

10.2 User needs translated into outputs

Operations users need comparisons across routes, stations and time periods. Pricing users need price and purchase comparisons. Customer-relations users need refund-request patterns linked to journey outcomes. The project team needs a traceable process, clear assumptions and reproducible calculations. These needs guide the dashboard structure and the acceptance checks used at the end of each phase.

11. User Stories and Use Cases

11.1 User stories

ID	As a…	I want to…	So that…
US-1	Data analyst	Clean and check the dataset and log every issue.	Results can be trusted.
US-2	Operations manager	See on-time, delayed and cancelled rates by route, station and time.	I can identify the weakest services.
US-3	Operations manager	See delay reasons ranked by frequency and delay minutes.	I can identify the largest recorded causes.
US-4	Pricing analyst	Compare prices by ticket type, class, railcard and booking lead time.	I can understand observed pricing patterns.
US-5	Pricing analyst	Compare sales by purchase channel and payment method.	I can understand how customers buy.
US-6	Customer-relations manager	See the share of delayed journeys with refund requests and how it varies.	I can understand observed claiming behaviour.
US-7	Station manager	Review a model that flags journeys at higher risk of delay.	I can assess whether the model offers useful advance information.
US-8	Any stakeholder	Use a dashboard with clear figures and filters.	I can explore results without technical support.

11.2 Use cases

ID / use case	Actor	Main steps
UC-1 — Prepare data	Data analyst	Load the file; confirm expected fields; correct date/time types; inspect duplicates, missing values and logic issues; create supported derived fields; save clean data and a quality log.
UC-2 — Analyse delays	Operations manager	Calculate status shares; compare by route, station, date and time; rank recorded delay reasons; review charts and tables.
UC-3 — Analyse pricing and purchases	Pricing analyst	Summarise prices by ticket type, class, railcard and route; compare with booking lead time; examine channels and payment methods.
UC-4 — Investigate refunds	Customer-relations manager	Calculate refund-request rates for a clearly defined group; compare by delay length, reason and ticket attributes; note that requests do not establish eligibility.
UC-5 — Predict delay risk	Data analyst	Choose a pre-journey target and valid inputs; split data into training and test sets; train a model; compare with a baseline; report metrics and important features.
UC-6 — View dashboard	Any stakeholder	Open the dashboard; review headline figures; apply filters; confirm that charts update to the selected subset.

11.3 Use-case acceptance

A use case is complete when its output can be reproduced, its key definitions are documented, and the results have been checked against the source data. For the dashboard, acceptance also requires readable titles, clear units, working filters and no unexplained differences between dashboard figures and report tables.

12. Functional Requirements

The requirements below consolidate the supplied requirements document. Priority labels are retained where stated. A requirement should be marked complete only after the team has evidence that it works.

ID	Area	The workflow shall…	Priority
FR-1	Data loading	Load the dataset and check that the expected 18 fields are present.	Must
FR-2	Data cleaning	Convert date/time values to suitable types, including arrivals after midnight.	Must
FR-3	Data quality	Detect duplicate Transaction IDs and report missing values, distinguishing expected blanks from errors.	Must
FR-4	Data quality	Check and log logic concerns such as arrival before departure, refund requests on on-time journeys and zero prices.	Must
FR-5	Standardisation	Standardise labels such as First and First Class and flag potential price or delay outliers.	Should
FR-6	Derived fields	Calculate delay minutes, booking lead time, route, day, hour and peak/off-peak flag where supported.	Must
FR-7	Descriptive analysis	Report summary statistics for price, delay minutes and booking lead time.	Must
FR-8	Journey status	Report on-time, delayed and cancelled shares by route, station, time, ticket type and class.	Must
FR-9	Delay analysis	Rank recorded delay reasons by frequency and delay minutes.	Must
FR-10	Pricing analysis	Compare price by ticket type, class, railcard and route, and against booking lead time.	Must
FR-11	Purchase analysis	Compare sales by purchase channel, payment method and purchase time.	Should
FR-12	Refund analysis	Report refund-request rates for defined delayed/cancelled groups by delay length and available attributes.	Must
FR-13	Statistical testing	Apply suitable tests, such as chi-square, t-test or ANOVA, when assumptions and data types support them.	Should
FR-14	Prediction	Build a baseline and at least one model to predict delayed versus on-time journeys using pre-journey information only.	Should
FR-15	Model evaluation	Use a training/test split and report accuracy, precision, recall, F1-score and important features where appropriate.	Should
FR-16	Dashboard KPIs	Show total journeys, on-time rate, average delay, total revenue and refund-request rate where fields support them.	Must
FR-17	Dashboard filters	Provide filters for date, route/station, ticket type, class, railcard, channel and journey status where available.	Should
FR-18	Visualisation	Produce clear charts with titles, labelled axes and an accessible colour palette.	Must
FR-19	Reporting	Export cleaned data and result tables and produce a report with findings, limitations and recommendations.	Must

12.1 Requirement traceability

The team should link each requirement to an implementation location and a test. This makes it easier to show that the work meets the agreed scope.

Requirement group	Evidence / output	Verification
FR-1 to FR-6	Cleaned dataset and quality log	Field check, before/after row counts, sample-record validation.
FR-7 to FR-13	Analysis tables and charts	Recalculate selected figures and verify group definitions.
FR-14 to FR-15	Model notebook or evaluation record	Check input features, data split, baseline comparison and metrics.
FR-16 to FR-18	Power BI dashboard	Test measures, filters, chart labels and refresh behaviour.
FR-19	Final report and exported outputs	Confirm findings match the dashboard and limitations are stated.

13. Non-functional Requirements

Non-functional requirements describe the quality of the workflow and its outputs rather than individual analytical features. Targets below follow the supplied requirements document and should be treated as testable goals.

Category	Requirement	How it will be measured
Performance	The workflow and dashboard should run quickly on a normal laptop.	Full pipeline under 5 minutes where feasible; dashboard filters refresh under 3 seconds.
Accuracy	Cleaning and calculated fields should be correct.	At least 30 records spot-checked against raw data with no unexplained errors.
Reliability	The workflow should stop with a clear message if data is not as expected.	Missing fields or invalid types trigger a documented warning or error.
Reproducibility	A team member should be able to repeat the analysis and obtain the same key results.	Documented steps, fixed random seeds where relevant and consistent outputs.
Usability	Results should be understandable to non-technical users.	Each chart has a title and a clear takeaway; a reviewer can use the dashboard without training.
Security and privacy	The project should not expose personal or payment-card details.	Only suitable aggregated results are shared; access to raw data is controlled.
Ethics	Railcard-group findings are descriptive and not used to profile individuals.	Group results are reported with appropriate caveats.
Model quality	Predictive models should be compared with a simple baseline on unseen data.	Held-out test metrics are reported; leakage fields are excluded.
Maintainability	The work should be documented so another person can follow it.	Field definitions, assumptions, calculations and file structure are recorded.

13.1 Usability and accessibility

- Use consistent terminology for on-time, delayed and cancelled journeys.
- Use readable font sizes and avoid relying on colour alone to communicate status.
- Label charts with units, dates and clear denominators where percentages are shown.
- Provide meaningful filter names and a clear way to return to an unfiltered view.
- Avoid overcrowding pages; prioritise the questions stakeholders need to answer.

13.2 Security, privacy and ethical interpretation

The dashboard should show only the detail needed for analysis. Personal information and payment-card details must not be exposed. Results involving railcard categories, including Disabled and Senior, should be described as aggregate patterns with caveats and should not be used to make claims about individual passengers. The team should also avoid causal language unless the analysis design supports causal conclusions.

14. Scope, Assumptions, and Limitations

14.1 Assumptions

- The dataset contains transaction-level UK railway ticket records with the fields described in the supplied documents.
- The actual dataset will be checked to confirm its structure, expected 18 fields and definitions.
- A recorded refund request is an observed action, not a verified compensation entitlement.
- The source data is sufficiently consistent to support at least descriptive analysis after documented cleaning.
- Predictive modelling is conditional on there being a clearly defined target, sufficient examples and suitable pre-journey inputs.

14.2 Limitations

- The dataset may not represent the whole UK railway network or all passenger groups.
- Missing or inconsistent fields can limit comparisons and may introduce bias.
- Delay reasons recorded in the data may not fully explain the operational cause of disruption.
- Predictive results depend on the selected period, data quality, target definition and features available before travel.
- A model's association between a field and delay does not by itself establish causation.
- No external weather or capacity data is assumed to be available in the current scope.

14.3 Final validation checklist

- The cleaned dataset has been saved separately from the raw file.
- The data-quality log includes major transformations and unresolved issues.
- KPI definitions, denominators and treatment of missing values are documented.
- Dashboard totals match the corresponding analysis outputs.
- Prediction inputs use only information available at the intended prediction time.
- Model results are compared with a baseline and reported with suitable metrics.
- The report distinguishes evidence from interpretation and includes limitations.
- Lecturer feedback and actual grading marks are added only when provided.

15. Conclusion

This document brings together the project plan, literature review and requirements for UK Railway Ticket Sales and Journey Performance Analysis. It defines a four-week delivery path, leaves team-member names blank for later assignment, identifies project risks and success measures, summarises relevant research, and sets out user needs and functional and non-functional requirements.

The project will be successful if it produces reliable and traceable analysis, a dashboard that stakeholders can understand, and conclusions that remain within the limits of the available evidence. A carefully documented workflow is as important as the final visuals: it allows the team to explain how the data was cleaned, how metrics were defined, why a model was selected and what the results can—and cannot—support.

Before submission, the team should confirm the dataset fields, fill in member names and course details, update the schedule to the actual course dates, and replace blank feedback and grading fields only with information supplied by the lecturer.

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